Department of Information and Computer Science

ICS445 – Network Management & Security

Phase 3 Project Report: Advanced Routing, Network Services, and Security Implementation

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Part 5 – Static and Dynamic Routing

In this phase, dynamic routing was established across the enterprise network using Single-Area OSPFv2 (Process ID 1, Area 0). Additionally, static default routing was configured at the network edge to provide internet access, complete with failover capabilities.

The configurations were applied and verified in the following topological order: Edge Router (R1), Core Switches (CSW), Distribution Switches (DSW), and Access Switches (ASW).

Edge Router (R1) Configuration

As the boundary device connecting the enterprise to the ISP, R1 handles both the OSPF backbone participation and the static default routing to the outside world.

- **Static Default Routing & Failover:** A primary static default route (0.0.0.0 0.0.0.0) was configured pointing to the primary ISP link. To ensure high availability, a floating static default route was configured pointing to the backup ISP link, using a higher Administrative Distance (AD) so it only becomes active if the primary link fails.
- **OSPF Configuration:** OSPF Process ID 1 was enabled. A loopback interface was used to manually set a stable Router ID.
- **Route Injection:** The *default-information originated* command was applied under the OSPF process to dynamically advertise the default route to all internal switches.
- **Passive Interfaces:** Interfaces connected to non-routing segments were set to passive to prevent unnecessary OSPF Hello packets and enhance security.

```
R1(config)#router ospf 1
R1(config-router)#router-id 10.0.0.76
R1(config-router)#default-information originate
R1(config-router)#passive-interface l0
R1(config-router)#int l0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#interface range g0/0-1
R1(config-if-range)#ip ospf 1 area 0
R1(config-if-range)#ip ospf network point-to-point
R1(config-if-range)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.1
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.5 2
R1(config)#do write
Building configuration...
[OK]
```

Figure 1. OSPF Process, Default Routing, and Route Injection Configuration on R1

Core Switches (CSW) Configuration

The core switches act as the high-speed routing backbone of the enterprise.

- **OSPF Activation:** OSPF Process 1 (Area 0) was enabled across all required Layer 3 interfaces (routed ports and SVIs) connecting the Core to R1 and the Distribution layer.
- **Router IDs:** Loopback interfaces were created and assigned as manual OSPF Router IDs for stability.
- **Network Types:** Where applicable, OSPF network types were explicitly configured (e.g., Point-to-Point on direct switch-to-router/switch-to-switch links) to bypass DR/BDR elections and speed up network convergence.

```
CSW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW1(config)#router ospf 1
CSW1(config-router)#passive-interface l0
CSW1(config-router)#interface range g1/0/1,g1/1/1-4
CSW1(config-if-range)#ip ospf network point-to-point
CSW1(config-if-range)#exit
CSW1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure.2 OSPF Configuration and Interface Settings on CSW1

```
CSW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW1(config)#router ospf 1
CSW1(config-router)#router-id 10.0.0.77
CSW1(config-router)#net 10.0.0.41 0.0.0.0 area 0
CSW1(config-router)#net 10.0.0.34 0.0.0.0 area 0
CSW1(config-router)#net 10.0.0.45 0.0.0.0 area 0
CSW1(config-router)#net 10.0.0.49 0.0.0.0 area 0
CSW1(config-router)#net 10.0.0.53 0.0.0.0 area 0
CSW1(config-router)#net 10.0.0.57 0.0.0.0 area 0
CSW1(config-router)#net 10.0.0.77 0.0.0.0 area 0
CSW1(config-router)#exit
CSW1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 3. OSPF Configuration and Interface Settings on CSW1

```
CSW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW2(config)#router ospf 1
CSW2(config-router)#passive-interface l0
CSW2(config-router)#interface range g1/0/1,g1/1/1-4
CSW2(config-if-range)#ip ospf network point-to-point
CSW2(config-if-range)#exit
CSW2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 4. OSPF Configuration and Interface Settings on CSW2

```
CSW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW2(config)#router ospf 1
CSW2(config-router)#router-id 10.0.0.78
CSW2(config-router)#net 10.0.0.42 0.0.0.0 area 0
CSW2(config-router)#net 10.0.0.38 0.0.0.0 area 0
CSW2(config-router)#net 10.0.0.61 0.0.0.0 area 0
CSW2(config-router)#net 10.0.0.65 0.0.0.0 area 0
CSW2(config-router)#net 10.0.0.69 0.0.0.0 area 0
CSW2(config-router)#net 10.0.0.73 0.0.0.0 area 0
CSW2(config-router)#net 10.0.0.78 0.0.0.0 area 0
CSW2(config-router)#exit
CSW2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 5. OSPF Configuration and Interface Settings on CSW2

Distribution Switches (DSW) Configuration

The distribution switches aggregate access-layer traffic and participate in the OSPF domain to route internal traffic to the Core.

- **OSPF Activation & Router IDs:** Similar to the core, OSPF Process 1 (Area 0) was enabled, and loopback addresses were used to manually configure the Router IDs.
- **Passive Interfaces:** This layer connects directly to user VLANs. All SVIs facing the access layer (where no OSPF routers exist) were configured as passive interfaces to optimize routing updates and secure the routing domain.

```
DSW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A1(config)#router ospf 1
DSW-A1(config-router)#passive-interface 10
DSW-A1(config-router)#passive-interface vlan 10
DSW-A1(config-router)#passive-interface vlan 20
DSW-A1(config-router)#interface range g1/1/1-2
DSW-A1(config-if-range)#ip ospf network point-to-point
DSW-A1(config-if-range)#exit
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A1(config)#router ospf 1
DSW-A1(config-router)#passive-interface vlan 40
DSW-A1(config-router)#exit
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A1(config)#router ospf 1
DSW-A1(config-router)#router-id 10.0.0.79
DSW-A1(config-router)#net 10.0.0.46 0.0.0.0 area 0
DSW-A1(config-router)#net 10.0.0.62 0.0.0.0 area 0
DSW-A1(config-router)#net 10.0.0.79 0.0.0.0 area 0
DSW-A1(config-router)#net 10.1.0.2 0.0.0.0 area 0
DSW-A1(config-router)#net 10.2.0.2 0.0.0.0 area 0
DSW-A1(config-router)#net 10.6.0.2 0.0.0.0 area 0
DSW-A1(config-router)#net 10.0.0.2 0.0.0.0 area 0
DSW-A1(config-router)#exit
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 6. OSPF and Passive Interface Configuration on DSWA1

```
DSW-A2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A2(config)#router ospf 1
DSW-A2(config-router)#passive-interface 10
DSW-A2(config-router)#passive-interface vlan 10
DSW-A2(config-router)#passive-interface vlan 20
DSW-A2(config-router)#interface range g1/1/1-2
DSW-A2(config-if-range)#ip ospf network point-to-point
DSW-A2(config-if-range)#exit
DSW-A2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A2(config)#
%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Speak -> Standby

DSW-A2(config)#router ospf 1
DSW-A2(config-router)#passive-interface vlan 40
DSW-A2(config-router)#exit
DSW-A2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A2(config)#
%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Standby -> Active

DSW-A2(config)#router ospf 1
DSW-A2(config-router)#router-id 10.0.0.80
DSW-A2(config-router)#net 10.0.0.50 0.0.0.0 area 0
DSW-A2(config-router)#net 10.0.0.66 0.0.0.0 area 0
DSW-A2(config-router)#net 10.0.0.80 0.0.0.0 area 0
DSW-A2(config-router)#net 10.1.0.3 0.0.0.0 area 0
DSW-A2(config-router)#net 10.2.0.3 0.0.0.0 area 0
DSW-A2(config-router)#net 10.6.0.3 0.0.0.0 area 0
DSW-A2(config-router)#net 10.0.0.3 0.0.0.0 area 0
DSW-A2(config-router)#exit
DSW-A2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 7. OSPF and Passive Interface Configuration on DSWA2

```
DSW-B1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-B1(config)#router ospf 1
DSW-B1(config-router)#passive-interface l0
DSW-B1(config-router)#passive-interface vlan 10
DSW-B1(config-router)#passive-interface vlan 20
DSW-B1(config-router)#interface range g1/1/1-2
DSW-B1(config-if-range)#ip ospf network point-to-point
DSW-B1(config-if-range)#exit
DSW-B1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-B1(config)#router ospf 1
DSW-B1(config-router)#passive-interface vlan 30
DSW-B1(config-router)#exit
DSW-B1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-B1(config)#router ospf 1
DSW-B1(config-router)#router-id 10.0.0.81
DSW-B1(config-router)#net 10.0.0.54 0.0.0.0 area 0
DSW-B1(config-router)#net 10.0.0.70 0.0.0.0 area 0
DSW-B1(config-router)#net 10.0.0.81 0.0.0.0 area 0
DSW-B1(config-router)#net 10.3.0.2 0.0.0.0 area 0
DSW-B1(config-router)#net 10.4.0.2 0.0.0.0 area 0
DSW-B1(config-router)#net 10.5.0.2 0.0.0.0 area 0
DSW-B1(config-router)#net 10.0.0.18 0.0.0.0 area 0
DSW-B1(config-router)#exit
DSW-B1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 8. OSPF and Passive Interface Configuration on DSWB1

Access Switches (ASW) Configuration

- **Configuration Note:** Access switches were configured to support the required VLANs and forward traffic to the Distribution layer. Since routing boundaries terminate at the Distribution layer SVIs, dynamic routing (OSPF) was not required at the Access layer.

Part 6 – Network Services

In this phase, critical network services were deployed to facilitate IP addressing, logging, management, and secure internet access. The configurations applied include DHCP, DNS, NTP, SNMP, Syslog, FTP file transfers, secure remote access (SSH), and Network Address Translation (NAT).

Edge Router (R1) Configuration

As the central router, R1 acts as the primary server for several services and handles internet translation.

- **DHCP Server:** Configured DHCP pools for the internal VLANs, explicitly excluding the first 10 usable IP addresses from each pool to reserve them for static assignment.
- **NTP Master:** Configured as a Stratum 5 NTP server, utilizing authentication (key 1) to securely synchronize time across the network.
- **NAT (Network Address Translation):** * Configured a Static NAT mapping for SRV1 to ensure it is reachable from the outside.
    - Configured Dynamic PAT using an Extended ACL (ACL 2) to identify internal traffic.
    - Configured the public NAT pool "POOL1" using the IP range 203.0.113.200 - 203.0.113.207 /29.
- **FTP & IOS Upgrade:** Utilized default FTP credentials to copy the updated IOS image from SRV1, followed by an automated reboot to load the new image.
- **Secure Management & Logging:** Generated RSA keys with the largest modulus, forced SSHv2-only access, applied an ACL to the VTY lines restricting access strictly to the Office A PCs subnet, configured an SNMP read-only community string, and directed 8192-byte buffered Syslog outputs to SRV1.

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip domain name jeremysitlab.com
R1(config)#ip name-server 10.5.0.4
R1(config)#ntp authentication-key 1 md5 ccna
R1(config)#ntp trusted-key 1
R1(config)#snmp-server community SNMPSTRING ro
%SNMP-5-WARMSTART: SNMP agent on host R1 is undergoing a warm start
R1(config)#logging 10.5.0.4
R1(config)#logging trap debugging
R1(config)#logging buffered 8192
R1(config)#crypto key generate rsa
The name for the keys will be: R1.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Feb 28 9:19:41.33: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#access-list 1 permit 10.1.0.0 0.0.0.255
R1(config)#line vty 0 15
R1(config-line)#access-class 1 in
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#logging synchronous
```

Figure 9. Extensive Service Configuration on R1 (DHCP, NTP, NAT, FTP, SSH, and Logging)

```
R1(config)#do write
Building configuration...
[OK]
R1(config)#ip dhcp excluded-address 10.0.0.1 10.0.0.10
R1(config)#ip dhcp excluded-address 10.1.0.1 10.0.0.10
R1(config)#ip dhcp excluded-address 10.2.0.1 10.0.0.10
R1(config)#ip dhcp excluded-address 10.0.0.17 10.0.0.26
R1(config)#ip dhcp excluded-address 10.3.0.1 10.0.0.10
R1(config)#ip dhcp excluded-address 10.4.0.1 10.0.0.10
R1(config)#ip dhcp excluded-address 10.6.0.1 10.0.0.10
R1(config)#ip dhcp pool A-Mgmt
R1(dhcp-config)#network 10.0.0.0 255.255.255.240
R1(dhcp-config)#default-router 10.0.0.1
R1(dhcp-config)#domain-name jeremysitlab.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#option 43 ip 10.0.0.7
R1(dhcp-config)#ip dhcp pool A-PC
R1(dhcp-config)#network 10.1.0.0 255.255.255.0
R1(dhcp-config)#default-router 10.1.0.1
R1(dhcp-config)#domain-name jeremysitlab.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#ip dhcp pool A-Phone
R1(dhcp-config)#network 10.2.0.0 255.255.255.0
R1(dhcp-config)#default-router 10.2.0.1
R1(dhcp-config)#domain-name jeremysitlab.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#ip dhcp pool B-Mgmt
R1(dhcp-config)#network 10.0.0.16 255.255.255.240
R1(dhcp-config)#default-router 10.0.0.17
R1(dhcp-config)#domain-name jeremysitlab.com
```

Figure 10. Extensive Service Configuration on R1 (DHCP, NTP, NAT, FTP, SSH, and Logging)

```
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#ip dhcp pool B-Mgmt
R1(dhcp-config)#network 10.0.0.16 255.255.255.240
R1(dhcp-config)#default-router 10.0.0.17
R1(dhcp-config)#domain-name jeremysitlab.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#option 43 ip 10.0.0.7
R1(dhcp-config)#ip dhcp pool B-PC
R1(dhcp-config)#network 10.3.0.0 255.255.255.0
R1(dhcp-config)#default-router 10.3.0.1
R1(dhcp-config)#domain-name jeremysitlab.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#ip dhcp pool B-Phone
R1(dhcp-config)#network 10.4.0.0 255.255.255.0
R1(dhcp-config)#default-router 10.4.0.1
R1(dhcp-config)#domain-name jeremysitlab.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#ip dhcp pool Wi-Fi
R1(dhcp-config)#network 10.6.0.0 255.255.255.0
R1(dhcp-config)#default-router 10.6.0.1
R1(dhcp-config)#domain-name jeremysitlab.com
R1(dhcp-config)#dns-server 10.5.0.4
R1(dhcp-config)#exit
R1(config)#ntp master 5
R1(config)#ntp server 216.239.35.0
R1(config)#ntp authentication-key 1 md5 ccna
R1(config)#ntp trusted-key 1
R1(config)#ip ftp username cisco
R1(config)#ip ftp password cisco
R1(config)#ip nat inside source static 10.5.0.4 203.0.113.113
```

Figure 11. Extensive Service Configuration on R1 (DHCP, NTP, NAT, FTP, SSH, and Logging)

```
R1(config)#interface range g0/0/0,g0/1/0
R1(config-if-range)#ip nat outside
R1(config-if-range)#interface range g0/0-1
R1(config-if-range)#ip nat inside
R1(config-if-range)#exit
R1(config)#access-list 2 permit 10.1.0.0 0.0.0.255
R1(config)#access-list 2 permit 10.2.0.0 0.0.0.255
R1(config)#access-list 2 permit 10.3.0.0 0.0.0.255
R1(config)#access-list 2 permit 10.4.0.0 0.0.0.255
R1(config)#access-list 2 permit 10.6.0.0 0.0.0.255
R1(config)#ip nat pool POOL1 203.0.113.200 203.0.113.207 netmask 255.255.255.248
R1(config)#ip nat inside source list 2 pool POOL1 overload
R1(config)#interface g0/0/0
R1(config-if)#shutdown

R1(config-if)#exit
R1(config)#router ospf 1
R1(config-router)#no default-information originate
R1(config-router)#default-information originate
R1(config-router)#exit
R1(config)#no cdp run
R1(config)#lldp run
R1(config)#do write
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to administratively down
```

Figure 12. Extensive Service Configuration on R1 (DHCP, NTP, NAT, FTP, SSH, and Logging)

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#do copy ftp flash
Address or name of remote host []? 10.5.0.4
Source filename []? c2900-universalk9-mz.SPA.155-3.M4a.bin
Destination filename [c2900-universalk9-mz.SPA.155-3.M4a.bin]?
```

Figure 13. Extensive Service Configuration on R1 (DHCP, NTP, NAT, FTP, SSH, and Logging)

```
R1#delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [c2900-universalk9-mz.SPA.151-4.M4.bin]?
Delete flash:/c2900-universalk9-mz.SPA.151-4.M4.bin? [confirm]
%Error deleting flash:/c2900-universalk9-mz.SPA.151-4.M4.bin (No such file or
directory)

R1#delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [c2900-universalk9-mz.SPA.151-4.M4.bin]?
Delete flash:/c2900-universalk9-mz.SPA.151-4.M4.bin? [confirm]
%Error deleting flash:/c2900-universalk9-mz.SPA.151-4.M4.bin (No such file or
directory)

R1#do write
      ^
% Invalid input detected at '^' marker.

R1#write
Building configuration...
```

Figure 14. Extensive Service Configuration on R1 (DHCP, NTP, NAT, FTP, SSH, and Logging)

Core Switches (CSW) Configuration

The core switches were configured to securely act as clients for the centralized management and time services.

- **NTP & DNS:** Configured to synchronize time with the R1 Loopback interface using NTP authentication (key 1) and utilize SRV1 for DNS resolution.
- **Secure Management & Logging:** Just like R1, generated high-modulus RSA keys, restricted VTY access to SSHv2 via an ACL allowing only Office A PCs, set the SNMP read-only community string, and forwarded buffered logs to SRV1.

```
CSW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW1(config)#ip domain name jeremysitlab.com
CSW1(config)#ip name-server 10.5.0.4
CSW1(config)#ntp authentication-key 1 md5 ccna
CSW1(config)#ntp trusted-key 1
CSW1(config)#snmp-server community SNMPSTRING ro
%SNMP-5-WARMSTART: SNMP agent on host CSW1 is undergoing a warm start
CSW1(config)#logging 10.5.0.4
CSW1(config)#logging trap debugging
CSW1(config)#logging buffered 8192
CSW1(config)#crypto key generate rsa
The name for the keys will be: CSW1.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

CSW1(config)#ip ssh version 2
*Feb 28 9:24:57.120: %SSH-5-ENABLED: SSH 1.99 has been enabled
CSW1(config)#access-list 1 permit 10.1.0.0 0.0.0.255
CSW1(config)#line vty 0 15
CSW1(config-line)#access-class 1 in
CSW1(config-line)#login local
CSW1(config-line)#transport input ssh
```

Figure 15. Network Services and Secure Access Configuration on CSW1

```
CSW1(config-line)#logging synchronous
CSW1(config-line)#exit
CSW1(config)#do write%SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disable

CSW1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
CSW1(config)#ntp server 10.0.0.76 key 1
CSW1(config)#no cdp run
CSW1(config)#lldp run
CSW1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
```

Figure 16. Network Services and Secure Access Configuration on CSW1

```
CSW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW2(config)#ip domain name jeremysitlab.com
CSW2(config)#ip name-server 10.5.0.4
CSW2(config)#ntp authentication-key 1 md5 ccna
CSW2(config)#ntp trusted-key 1
CSW2(config)#snmp-server community SNMPSTRING ro
%SNMP-5-WARMSTART: SNMP agent on host CSW2 is undergoing a warm start
CSW2(config)#logging 10.5.0.4
CSW2(config)#logging trap debugging
CSW2(config)#logging buffered 8192
CSW2(config)#crypto key generate rsa
The name for the keys will be: CSW2.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

CSW2(config)#ip ssh version 2
*Feb 28 9:26:42.417: %SSH-5-ENABLED: SSH 1.99 has been enabled
CSW2(config)#access-list 1 permit 10.1.0.0 0.0.0.255
CSW2(config)#line vty 0 15
CSW2(config-line)#access-class 1 in
CSW2(config-line)#login local
CSW2(config-line)#transport input ssh
CSW2(config-line)#logging synchronous
CSW2(config-line)#exit
CSW2(config)#do write%SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disabled, filtering disabled, size
```

Figure 17. Network Services and Secure Access Configuration on CSW2

```
CSW2(config)#ntp server 10.0.0.76 key 1
CSW2(config)#no cdp run
CSW2(config)#lldp run
CSW2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 18. Network Services and Secure Access Configuration on CSW2

Distribution Switches (DSW) Configuration

The distribution switches act as the gateway for end-users and require specific relay configurations alongside the standard management baseline.

- **DHCP Relay:** The ip helper-address command was applied to the necessary VLAN SVIs to forward client DHCP broadcast requests across the routed network to the DHCP pools hosted on R1.
- **Standard Services:** Configured NTP targeting the R1 Loopback (with authentication), DNS pointing to SRV1, SNMP read-only community strings, and Syslog forwarding to SRV1.
- **Secure Management:** Enforced SSHv2 on VTY lines secured by an ACL, allowing remote administration solely from the Office A PCs subnet.

```
DSW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A1(config)#ip domain name jeremysitlab.com
DSW-A1(config)#ip name-server 10.5.0.4
DSW-A1(config)#ntp authentication-key 1 md5 ccna
DSW-A1(config)#ntp trusted-key 1
DSW-A1(config)#snmp-server community SNMPSTRING ro
%SNMP-5-WARMSTART: SNMP agent on host DSW-A1 is undergoing a warm start
DSW-A1(config)#logging 10.5.0.4
DSW-A1(config)#logging trap debugging
DSW-A1(config)#logging buffered 8192
DSW-A1(config)#crypto key generate rsa
The name for the keys will be: DSW-A1.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

DSW-A1(config)#ip ssh version 2
*Feb 28 9:29:14.933: %SSH-5-ENABLED: SSH 1.99 has been enabled
DSW-A1(config)#access-list 1 permit 10.1.0.0 0.0.0.255
DSW-A1(config)#line vty 0 15
DSW-A1(config-line)#access-class 1 in
DSW-A1(config-line)#login local
DSW-A1(config-line)#transport input ssh
DSW-A1(config-line)#logging synchronous
DSW-A1(config-line)#exit
```

Figure 19. DHCP Relay and Services Configuration on DSWA1

```
DSW-A1(config)#interface vlan 10
DSW-A1(config-if)#ip helper-address 10.0.0.76
DSW-A1(config-if)#interface vlan 20
DSW-A1(config-if)#ip helper-address 10.0.0.76
DSW-A1(config-if)#interface vlan 99
DSW-A1(config-if)#ip helper-address 10.0.0.76
DSW-A1(config-if)#exit
DSW-A1(config)#ntp server 10.0.0.76 key 1
DSW-A1(config)#no cdp run
DSW-A1(config)#lldp run
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A1(config)#interface vlan 40
DSW-A1(config-if)#ip helper-address 10.0.0.76
DSW-A1(config-if)#exit
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 20. DHCP Relay and Services Configuration on DSWA1

```
DSW-A2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A2(config)#ip domain name jeremysitlab.com
DSW-A2(config)#ip name-server 10.5.0.4
DSW-A2(config)#ntp authentication-key 1 md5 ccna
DSW-A2(config)#ntp trusted-key 1
DSW-A2(config)#snmp-server community SNMPSTRING ro
%SNMP-5-WARMSTART: SNMP agent on host DSW-A2 is undergoing a warm start
DSW-A2(config)#logging 10.5.0.4
DSW-A2(config)#logging trap debugging
DSW-A2(config)#logging buffered 8192
DSW-A2(config)#crypto key generate rsa
The name for the keys will be: DSW-A2.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

DSW-A2(config)#ip ssh version 2
*Feb 28 9:31:3.396: %SSH-5-ENABLED: SSH 1.99 has been enabled
DSW-A2(config)#access-list 1 permit 10.1.0.0 0.0.0.255
DSW-A2(config)#line vty 0 15
DSW-A2(config-line)#access-class 1 in
DSW-A2(config-line)#login local
DSW-A2(config-line)#transport input ssh
DSW-A2(config-line)#logging synchronous
DSW-A2(config-line)#exit
```

Figure 21. DHCP Relay and Services Configuration on DSWA2

```
DSW-A2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A2(config)#interface vlan 10
DSW-A2(config-if)#ip helper-address 10.0.0.76
DSW-A2(config-if)#interface vlan 20
DSW-A2(config-if)#ip helper-address 10.0.0.76
DSW-A2(config-if)#interface vlan 99
DSW-A2(config-if)#ip helper-address 10.0.0.76
DSW-A2(config-if)#exit
DSW-A2(config)#ntp server 10.0.0.76 key 1
DSW-A2(config)#no cdp run
DSW-A2(config)#lldp run
DSW-A2(config)#do write
%LINK-5-CHANGED: Interface Vlan99, changed state to up
```

Figure 22. DHCP Relay and Services Configuration on DSWA2

```
DSW-A2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A2(config)#interface vlan 40
DSW-A2(config-if)#ip helper-address 10.0.0.76
DSW-A2(config-if)#exit
DSW-A2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 23. DHCP Relay and Services Configuration on DSWA2

```
DSW-B1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-B1(config)#ip domain name jeremysitlab.com
DSW-B1(config)#ip name-server 10.5.0.4
DSW-B1(config)#ntp authentication-key 1 md5 ccna
DSW-B1(config)#ntp trusted-key 1
DSW-B1(config)#snmp-server community SNMPSTRING ro
%SNMP-5-WARMSTART: SNMP agent on host DSW-B1 is undergoing a warm start
DSW-B1(config)#logging 10.5.0.4
DSW-B1(config)#logging trap debugging
DSW-B1(config)#logging buffered 8192
DSW-B1(config)#crypto key generate rsa
The name for the keys will be: DSW-B1.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

DSW-B1(config)#ip ssh version 2
*Feb 28 9:34:35.287: %SSH-5-ENABLED: SSH 1.99 has been enabled
DSW-B1(config)#access-list 1 permit 10.1.0.0 0.0.0.255
DSW-B1(config)#line vty 0 15
DSW-B1(config-line)#access-class 1 in
DSW-B1(config-line)#login local
DSW-B1(config-line)#transport input ssh
DSW-B1(config-line)#logging synchronous
DSW-B1(config-line)#exit
```

Figure 24. DHCP Relay and Services Configuration on DSWB1

```
DSW-B1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-B1(config)#interface vlan 10
DSW-B1(config-if)#ip helper-address 10.0.0.76
DSW-B1(config-if)#interface vlan 20
DSW-B1(config-if)#ip helper-address 10.0.0.76
DSW-B1(config-if)#interface vlan 99
DSW-B1(config-if)#ip helper-address 10.0.0.76
DSW-B1(config-if)#exit
DSW-B1(config)#ntp server 10.0.0.76 key 1
DSW-B1(config)#no cdp run
DSW-B1(config)#lldp run
DSW-B1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-B1(config)#interface vlan 30
DSW-B1(config-if)#ip helper-address 10.0.0.76
DSW-B1(config-if)#exit
DSW-B1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 25. DHCP Relay and Services Configuration on DSWB1

```
DSW-B2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-B2(config)#ip domain name jeremysitlab.com
DSW-B2(config)#ip name-server 10.5.0.4
DSW-B2(config)#ntp authentication-key 1 md5 ccna
DSW-B2(config)#ntp trusted-key 1
DSW-B2(config)#snmp-server community SNMPSTRING ro
%SNMP-5-WARMSTART: SNMP agent on host DSW-B2 is undergoing a warm start
DSW-B2(config)#logging 10.5.0.4
DSW-B2(config)#logging trap debugging
DSW-B2(config)#logging buffered 8192
DSW-B2(config)#crypto key generate rsa
The name for the keys will be: DSW-B2.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]
```

Figure 26. DHCP Relay and Services Configuration on DSWB2

```
DSW-B2(config)#interface vlan 10
DSW-B2(config-if)#ip helper-address 10.0.0.76
DSW-B2(config-if)#interface vlan 20
DSW-B2(config-if)#ip helper-address 10.0.0.76
DSW-B2(config-if)#interface vlan 99
DSW-B2(config-if)#ip helper-address 10.0.0.76
DSW-B2(config-if)#exit
DSW-B2(config)#ntp server 10.0.0.76 key 1
DSW-B2(config)#no cdp run
DSW-B2(config)#lldp run
DSW-B2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-B2(config)#interface vlan 30
DSW-B2(config-if)#ip helper-address 10.0.0.76
DSW-B2(config-if)#exit
DSW-B2(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Figure 27. DHCP Relay and Services Configuration on DSWB2

Access Switches (ASW) Configuration

The access layer devices received the standardized management baseline to ensure uniform security and monitoring across the topology.

- **Standard Services:** Configured NTP synchronization with R1, DNS resolution via SRV1, SNMP read-only strings, and Syslog message generation to SRV1.
- **Secure Management:** Restricted remote management to SSHv2, applying an ACL on the VTY lines to restrict access to the designated Office A PC subnet.

```
ASW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A1(config)#ip domain name jeremysitlab.com
ASW-A1(config)#ip name-server 10.5.0.4
ASW-A1(config)#ntp authentication-key 1 md5 ccna
ASW-A1(config)#ntp trusted-key 1
ASW-A1(config)#snmp-server community SNMPSTRING ro
ASW-A1(config)#logging 10.5.0.4
ASW-A1(config)#logging trap debugging
ASW-A1(config)#logging buffered 8192
ASW-A1(config)#crypto key generate rsa
The name for the keys will be: ASW-A1.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

ASW-A1(config)#ip ssh version 2
*Feb 28 9:38:40.997: %SSH-5-ENABLED: SSH 1.99 has been enabled
ASW-A1(config)#access-list 1 permit 10.1.0.0 0.0.0.255
ASW-A1(config)#line vty 0 15
ASW-A1(config-line)#access-class 1 in
ASW-A1(config-line)#login local
ASW-A1(config-line)#transport input ssh
ASW-A1(config-line)#logging synchronous
ASW-A1(config-line)#exit
```

Figure 28. Network Services Configuration on ASWA1

```
ASW-A1(config)#do write
Building configuration...
[OK]
ASW-A1(config)#ntp server 10.0.0.76 key 1
ASW-A1(config)#no cdp run
ASW-A1(config)#lldp run
ASW-A1(config)#interface f0/1
ASW-A1(config-if)#no lldp transmit
ASW-A1(config-if)#exit
ASW-A1(config)#do write
Building configuration...
[OK]
ASW-A1(config)#
```

Figure 29. Network Services Configuration on ASWA1

```
ASW-A2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A2(config)#ip domain name jeremysitlab.com
ASW-A2(config)#ip name-server 10.5.0.4
ASW-A2(config)#ntp authentication-key 1 md5 ccna
ASW-A2(config)#ntp trusted-key 1
ASW-A2(config)#snmp-server community SNMPSTRING ro
ASW-A2(config)#logging 10.5.0.4
ASW-A2(config)#logging trap debugging
ASW-A2(config)#logging buffered 8192
ASW-A2(config)#crypto key generate rsa
The name for the keys will be: ASW-A2.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

ASW-A2(config)#ip ssh version 2
*Feb 28 9:39:51.453: %SSH-5-ENABLED: SSH 1.99 has been enabled
ASW-A2(config)#access-list 1 permit 10.1.0.0 0.0.0.255
ASW-A2(config)#line vty 0 15
ASW-A2(config-line)#access-class 1 in
ASW-A2(config-line)#login local
ASW-A2(config-line)#transport input ssh
ASW-A2(config-line)#logging synchronous
ASW-A2(config-line)#exit
```

Figure 30. Network Services Configuration on ASWA2

```
ASW-A2(config)#do write
Building configuration...
[OK]
ASW-A2(config)#ntp server 10.0.0.76 key 1
ASW-A2(config)#no cdp run
ASW-A2(config)#lldp run
ASW-A2(config)#interface f0/1
ASW-A2(config-if)#no lldp transmit
ASW-A2(config-if)#exit
ASW-A2(config)#do write
Building configuration...
[OK]
```

Figure 31. Network Services Configuration on ASWA2

```
ASW-A3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A3(config)#ip domain name jeremysitlab.com
ASW-A3(config)#ip name-server 10.5.0.4
ASW-A3(config)#ntp authentication-key 1 md5 ccna
ASW-A3(config)#ntp trusted-key 1
ASW-A3(config)#snmp-server community SNMPSTRING ro
ASW-A3(config)#logging 10.5.0.4
ASW-A3(config)#logging trap debugging
ASW-A3(config)#logging buffered 8192
ASW-A3(config)#crypto key generate rsa
The name for the keys will be: ASW-A3.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

ASW-A3(config)#ip ssh version 2
*Feb 28 9:41:8.851: %SSH-5-ENABLED: SSH 1.99 has been enabled
ASW-A3(config)#access-list 1 permit 10.1.0.0 0.0.0.255
ASW-A3(config)#line vty 0 15
ASW-A3(config-line)#access-class 1 in
ASW-A3(config-line)#login local
ASW-A3(config-line)#transport input ssh
ASW-A3(config-line)#logging synchronous
ASW-A3(config-line)#exit
ASW-A3(config)#do write%SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugg
```

Figure 32. Network Services Configuration on ASWA3

```
ASW-A3(config)#do write
Building configuration...
[OK]
ASW-A3(config)#ntp server 10.0.0.76 key 1
ASW-A3(config)#no cdp run
ASW-A3(config)#lldp run
ASW-A3(config)#interface f0/1
ASW-A3(config-if)#no lldp transmit
ASW-A3(config-if)#exit
ASW-A3(config)#do write
Building configuration...
[OK]
```

Figure 33. Network Services Configuration on ASWA3

```
ASW-B1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B1(config)#ip domain name jeremysitlab.com
ASW-B1(config)#ip name-server 10.5.0.4
ASW-B1(config)#ntp authentication-key 1 md5 ccna
ASW-B1(config)#ntp trusted-key 1
ASW-B1(config)#snmp-server community SNMPSTRING ro
ASW-B1(config)#logging 10.5.0.4
ASW-B1(config)#logging trap debugging
ASW-B1(config)#logging buffered 8192
ASW-B1(config)#crypto key generate rsa
The name for the keys will be: ASW-B1.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

ASW-B1(config)#ip ssh version 2
*Feb 28 9:42:21.552: %SSH-5-ENABLED: SSH 1.99 has been enabled
ASW-B1(config)#access-list 1 permit 10.1.0.0 0.0.0.255
ASW-B1(config)#line vty 0 15
ASW-B1(config-line)#access-class 1 in
ASW-B1(config-line)#login local
ASW-B1(config-line)#transport input ssh
ASW-B1(config-line)#logging synchronous
ASW-B1(config-line)#exit
```

Figure 34. Network Services Configuration on ASWB1

```
ASW-B1(config)#do write
Building configuration...
[OK]
ASW-B1(config)#ntp server 10.0.0.76 key 1
ASW-B1(config)#no cdp run
ASW-B1(config)#lldp run
ASW-B1(config)#interface f0/1
ASW-B1(config-if)#no lldp transmit
ASW-B1(config-if)#exit
ASW-B1(config)#do write
Building configuration...
[OK]
```

Figure 35. Network Services Configuration on ASWB1

```
ASW-B2(config)#do write
Building configuration...
[OK]
ASW-B2(config)#ntp server 10.0.0.76 key 1
ASW-B2(config)#no cdp run
ASW-B2(config)#lldp run
ASW-B2(config)#interface f0/1
ASW-B2(config-if)#no lldp transmit
ASW-B2(config-if)#exit
ASW-B2(config)#do write
Building configuration...
[OK]
```

Figure 36. Network Services Configuration on ASWB2

```
ASW-B2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B2(config)#ip domain name jeremysitlab.com
ASW-B2(config)#ip name-server 10.5.0.4
ASW-B2(config)#ntp authentication-key 1 md5 ccna
ASW-B2(config)#ntp trusted-key 1
ASW-B2(config)#snmp-server community SNMPSTRING ro
ASW-B2(config)#logging 10.5.0.4
ASW-B2(config)#logging trap debugging
ASW-B2(config)#logging buffered 8192
ASW-B2(config)#crypto key generate rsa
The name for the keys will be: ASW-B2.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]

ASW-B2(config)#ip ssh version 2
*Feb 28 9:44:19.313: %SSH-5-ENABLED: SSH 1.99 has been enabled
ASW-B2(config)#access-list 1 permit 10.1.0.0 0.0.0.255
ASW-B2(config)#line vty 0 15
ASW-B2(config-line)#access-class 1 in
ASW-B2(config-line)#login local
ASW-B2(config-line)#transport input ssh
ASW-B2(config-line)#logging synchronous
ASW-B2(config-line)#exit
ASW-B2(config)#do write%SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debuggi
```

Figure 37. Network Services Configuration on ASWB2

```
ASW-B3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B3(config)#ip domain name jeremysitlab.com
ASW-B3(config)#ip name-server 10.5.0.4
ASW-B3(config)#ntp authentication-key 1 md5 ccna
ASW-B3(config)#ntp trusted-key 1
ASW-B3(config)#snmp-server community SNMPSTRING ro
ASW-B3(config)#logging 10.5.0.4
ASW-B3(config)#logging trap debugging
ASW-B3(config)#logging buffered 8192
ASW-B3(config)#crypto key generate rsa
The name for the keys will be: ASW-B3.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 4096
% Generating 4096 bit RSA keys, keys will be non-exportable...[OK]
```

Figure 38. Network Services Configuration on ASWB3

```
ASW-B3(config)#do write
Building configuration...
[OK]
ASW-B3(config)#ntp server 10.0.0.76 key 1
ASW-B3(config)#no cdp run
ASW-B3(config)#lldp run
ASW-B3(config)#interface f0/1
ASW-B3(config-if)#no lldp transmit
ASW-B3(config-if)#exit
ASW-B3(config)#do write
Building configuration...
[OK]
```

Figure 39. Network Services Configuration on ASWB3

Verification and Testing

To confirm the services were operational from an end-user perspective, tests were conducted from an Office A endpoint (PC1).

1. **IP Addressing:** The output of ipconfig confirmed the PC successfully obtained an IP address, Subnet Mask, Default Gateway, and DNS Server address dynamically via the DSW DHCP relay to R1.
2. **Remote Access:** SSH access was successfully initiated from PC1 to the network devices, proving that VTY ACL correctly permitted the Office A subnet.
3. **Connectivity & NAT:** ICMP Echo Requests (Pings) were sent to external destinations, confirming that the dynamic PAT translations on R1 were functioning properly and routing Internet traffic.

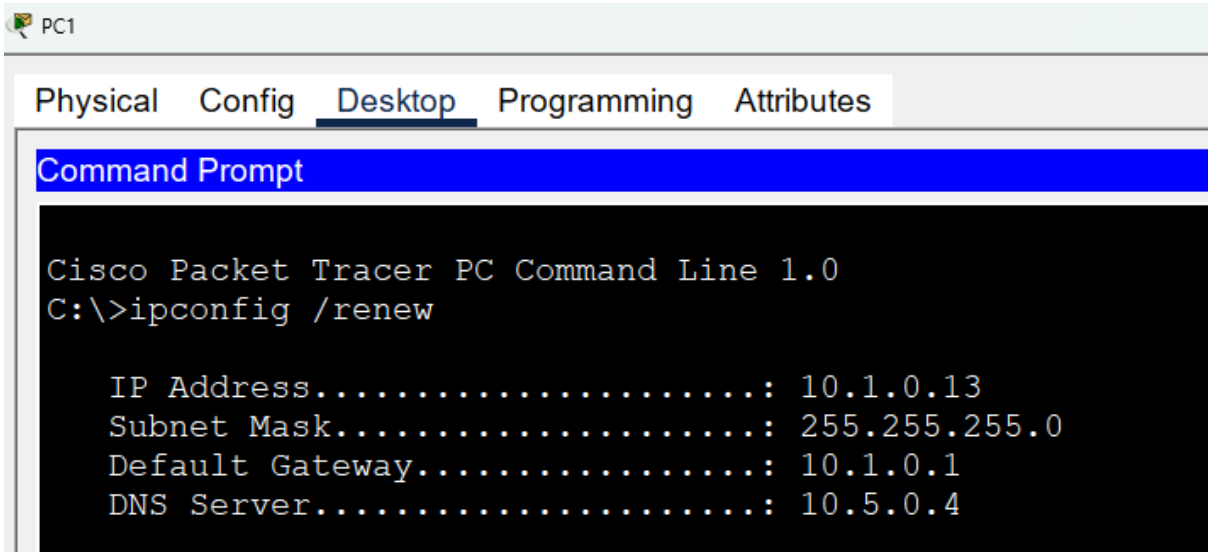

Figure 40. DHCP Address Verification on PC1

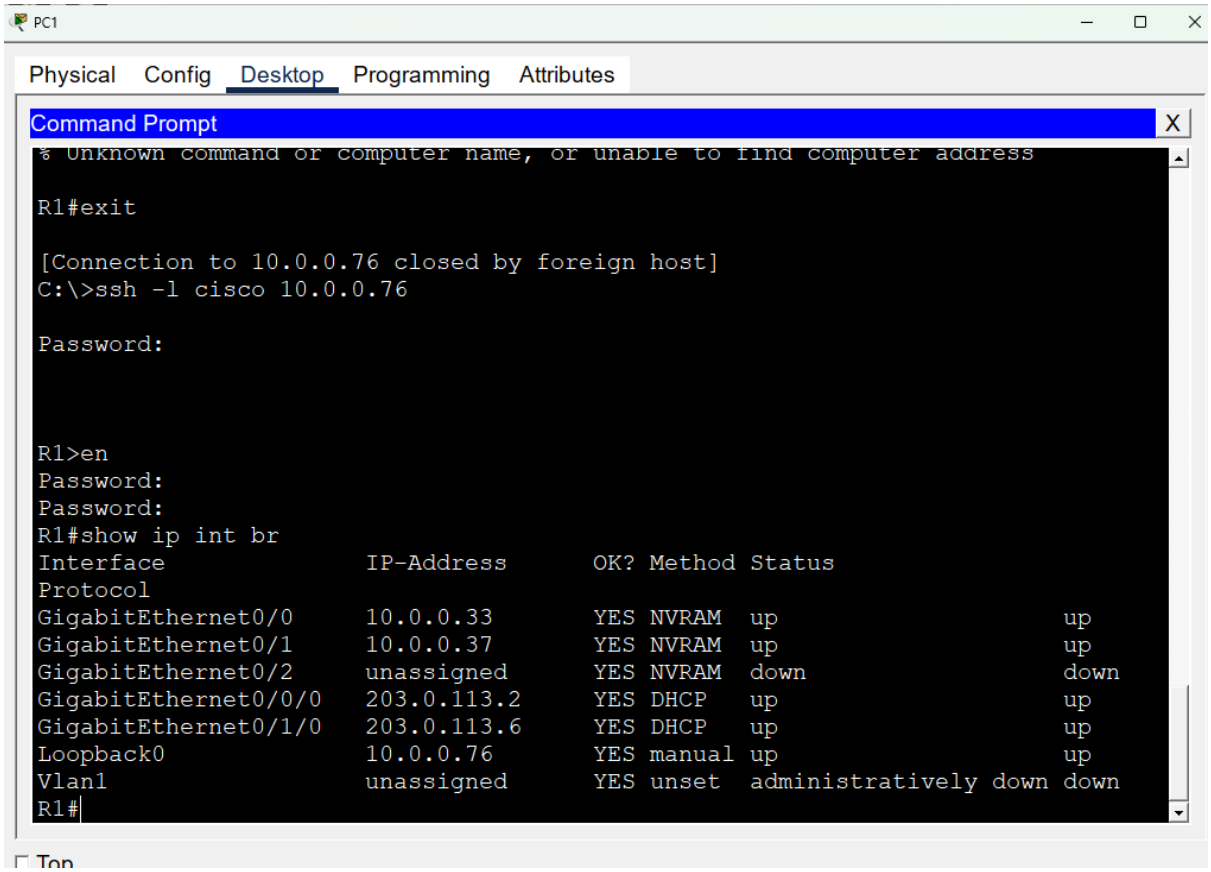

Figure 41. Successful SSH Remote Access from PC1

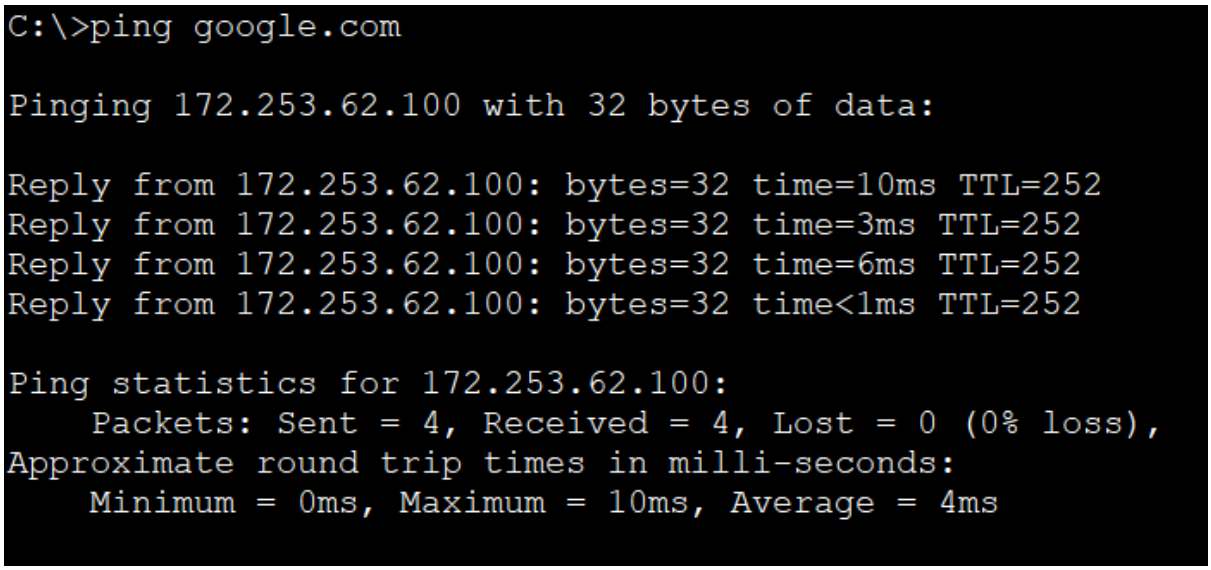

Figure 42. Connectivity and NAT Verification from PC1

Part 7 – Security Features

Extended ACL

- **Configure OfficeA_to_OfficeB**
  As shown in Figure 43, an extended ACL named OfficeA_to_OfficeB was configured to control traffic between Office A and Office B. It permits ICMP, denies other IP traffic between the two networks, and allows all remaining traffic.

```
DSW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A1(config)#do show acce
DSW-A1(config)#do show acces
DSW-A1(config)#do show access-list
Standard IP access list 1
    10 permit 10.1.0.0 0.0.0.255
Extended IP access list OfficeA_to_OfficeB
    10 permit icmp 10.1.0.0 0.0.0.255 10.3.0.0 0.0.0.255
    20 deny ip 10.1.0.0 0.0.0.255 10.3.0.0 0.0.0.255
    30 permit ip any any (584 match(es))
```

Figure 43. Extended ACL OfficeA_to_OfficeB configuration and verification

- **Apply using extended ACL best practice placement**
  The extended ACL was placed close to the source network, following best practice for extended ACL deployment. As shown in Figure 44, the ACL verification on CSW1, as illustrated in Figure 45, the ACL verification on R1.

```
CSW1(config)#do show access-list
Standard IP access list 1
    10 permit 10.1.0.0 0.0.0.255

CSW1(config)#
```

Figure 44. ACL verification on CSW1

```
R1(config)#do show access-list
Standard IP access list 1
    10 permit 10.1.0.0 0.0.0.255 (4 match(es))
Standard IP access list 2
    10 permit 10.1.0.0 0.0.0.255 (10 match(es))
    20 permit 10.2.0.0 0.0.0.255
    30 permit 10.3.0.0 0.0.0.255
    40 permit 10.4.0.0 0.0.0.255
    50 permit 10.6.0.0 0.0.0.255
```

Figure 45. ACL verification on R1

Port Security

As shown in Figure 46,

- **Configure on access ports**
  Port security was enabled on access ports to prevent unauthorized devices from connecting to the network. Port security was enabled on access port Fa0/1.
- **Set minimum required MAC addresses**
  The maximum number of secure MAC addresses was set to 2, allowing only the required devices on the access port.
- **Configure violation mode to block invalid traffic**
  Violation mode was configured as restricted to block unauthorized traffic without shutting down the port.
- **Save learned MAC addresses**
  Sticky MAC learning was enabled so the switch could automatically learn and save secure MAC addresses.
- **Verify port security**
  The port security settings were verified using show port-security interface fa0/1.

```
ASW-A2(config)#do show port-security interface f0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0001.4290.A4D1:10
Security Violation Count   : 0
```

Figure 46. Port security verification on Fa0/1

DHCP Snooping

As shown in Figure 47,

- **Enable on all active VLANs**
  DHCP snooping was enabled on VLANs 10, 20, 40, and 99.
- **Configure trusted ports**
  The uplink interfaces GigabitEthernet0/1 and GigabitEthernet0/2 were configured as trusted ports.
- **Disable Option 82 insertion**
  Option 82 insertion was disabled in the DHCP snooping configuration.
- **Configure rate limits**

Rate limits were applied on untrusted access ports to protect against DHCP starvation attacks.

- **Verify DHCP snooping**
  The DHCP snooping configuration was verified using show ip dhcp snooping.

```
DHCP snooping is configured on following VLANs:
10,20,40,99
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                  Trusted    Rate limit (pps)
-----------------------    -------    ----------------
FastEthernet0/2            no         100
FastEthernet0/3            no         unlimited
FastEthernet0/1            no         15
FastEthernet0/4            no         unlimited
FastEthernet0/6            no         unlimited
FastEthernet0/5            no         unlimited
FastEthernet0/7            no         unlimited
FastEthernet0/9            no         unlimited
FastEthernet0/8            no         unlimited
FastEthernet0/12           no         unlimited
FastEthernet0/10           no         unlimited
FastEthernet0/14           no         unlimited
FastEthernet0/11           no         unlimited
FastEthernet0/13           no         unlimited
FastEthernet0/17           no         unlimited
FastEthernet0/18           no         unlimited
FastEthernet0/15           no         unlimited
FastEthernet0/16           no         unlimited
FastEthernet0/21           no         unlimited
FastEthernet0/23           no         unlimited
FastEthernet0/19           no         unlimited
FastEthernet0/20           no         unlimited
FastEthernet0/22           no         unlimited
FastEthernet0/24           no         unlimited
GigabitEthernet0/1         yes        unlimited
GigabitEthernet0/2         yes        unlimited
ASW-A1(config)#
ASW-A1(config)#
```

Figure 47. DHCP snooping configuration and verification

Dynamic ARP Inspection (DAI)

As shown Figure 48,

- **Enable on all active VLANs**
  Dynamic ARP Inspection was enabled on VLANs 10, 20, 40, and 99.
- **Configure trusted ports**
  The appropriate uplink interfaces were configured as trusted ports for ARP inspection.
- **Enable validation checks**
  ARP validation checks for source MAC address, destination MAC address, and IP address were enabled.
- **Verify DAI**
  The DAI configuration was verified using "do show ip arp inspection" command.

```
ASW-A1(config)#do show ip arp inspection

Source Mac Validation      : Enabled
Destination Mac Validation : Enabled
IP Address Validation      : Enabled

 Vlan     Configuration    Operation   ACL Match          Static ACL
 ----     -------------    ---------   ---------          ----------
   10     Enabled          Active
   20     Enabled          Active
   40     Enabled          Active
   99     Enabled          Active

 Vlan     ACL Logging      DHCP Logging      Probe Logging
 ----     -----------      ------------      -------------
   10     Deny             Deny              Off
   20     Deny             Deny              Off
   40     Deny             Deny              Off
   99     Deny             Deny              Off

 Vlan      Forwarded        Dropped     DHCP Drops      ACL Drops
 ----      ---------        -------     ----------      ---------
   10              0              0              0              0
 --More--
```

Figure 48. Dynamic ARP Inspection configuration and verification

Part 8 – IPv6

R1

IPv6 routing was enabled on R1 using ipv6 unicast-routing, as shown in Figure 49. IPv6 addresses were then configured on the R1 interfaces to support the connected IPv6 networks, as shown in Figure 50. In addition, two IPv6 default static routes were configured on R1: a recursive route using the next-hop address and a floating route using the exit interface, next-hop address, and a higher administrative distance, as shown in Figure 49. The IPv6 routing table was checked to verify the configured routes, as shown in Figure 51.

```
R1(config)#do show run | inc ipv6
ipv6 unicast-routing
no ipv6 cef
 ipv6 address 2001:DB8:A1::/64 eui-64
 ipv6 address 2001:DB8:A2::/64 eui-64
 ipv6 address 2001:DB8:A::2/64
 ipv6 address 2001:DB8:B::2/64
ipv6 route ::/0 GigabitEthernet0/1/0 2001:DB8:B::1 2
ipv6 route ::/0 2001:DB8:A::1
```

Figure 49. IPv6 routing and default static route configuration on R1

```
R1(config)#do show ipv6 int br
GigabitEthernet0/0         [up/up]
    FE80::20C:CFFF:FE08:5001
    2001:DB8:A1:0:20C:CFFF:FE08:5001
GigabitEthernet0/1         [up/up]
    FE80::20C:CFFF:FE08:5002
    2001:DB8:A2:0:20C:CFFF:FE08:5002
GigabitEthernet0/2         [up/down]
    unassigned
GigabitEthernet0/0/0       [up/up]
    FE80::2E0:A3FF:FE95:4D4E
    2001:DB8:A::2
GigabitEthernet0/1/0       [up/up]
    FE80::2E0:F7FF:FE68:6426
    2001:DB8:B::2
Loopback0                  [up/up]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
```

Figure 50. IPv6 addresses configured on R1 interfaces

```
R1(config)#do show ipv6 route
IPv6 Routing Table - 10 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   ::/0 [1/0]
     via 2001:DB8:A::1
C   2001:DB8:A::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:A::2/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:DB8:B::/64 [0/0]
     via GigabitEthernet0/1/0, directly connected
L   2001:DB8:B::2/128 [0/0]
     via GigabitEthernet0/1/0, receive
C   2001:DB8:A1::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:A1:0:20C:CFFF:FE08:5001/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:DB8:A2::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:A2:0:20C:CFFF:FE08:5002/128 [0/0]
     via GigabitEthernet0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

Figure 51. IPv6 routing table verification on R1

CSW1

An IPv6 address was configured on CSW1 to enable IPv6 connectivity on the switch interface, as shown in Figure 52.

```
CSW1(config)#do show ipv6 int g1/0/1
GigabitEthernet1/0/1 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::20D:BDFF:FEEB:5E01
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:A1:0:20D:BDFF:FEEB:5E01, subnet is 2001:DB8:A1::/64 [EUI]
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FFEB:5E01
```

Figure 52. IPv6 address configured on CSW1

CSW2

An IPv6 address was also configured on CSW2 to support IPv6 connectivity within the network.

Part 9 – Wireless

WLC1 Access and Dynamic Interface Configuration

WLC1 was accessed through the GUI, and the dynamic interface for the Wi-Fi VLAN was configured, as shown in Figure 53. The interface Wi-Fi was assigned to VLAN 40 with IP address 10.6.0.2, subnet mask 255.255.255.0, and default gateway 10.6.0.1.

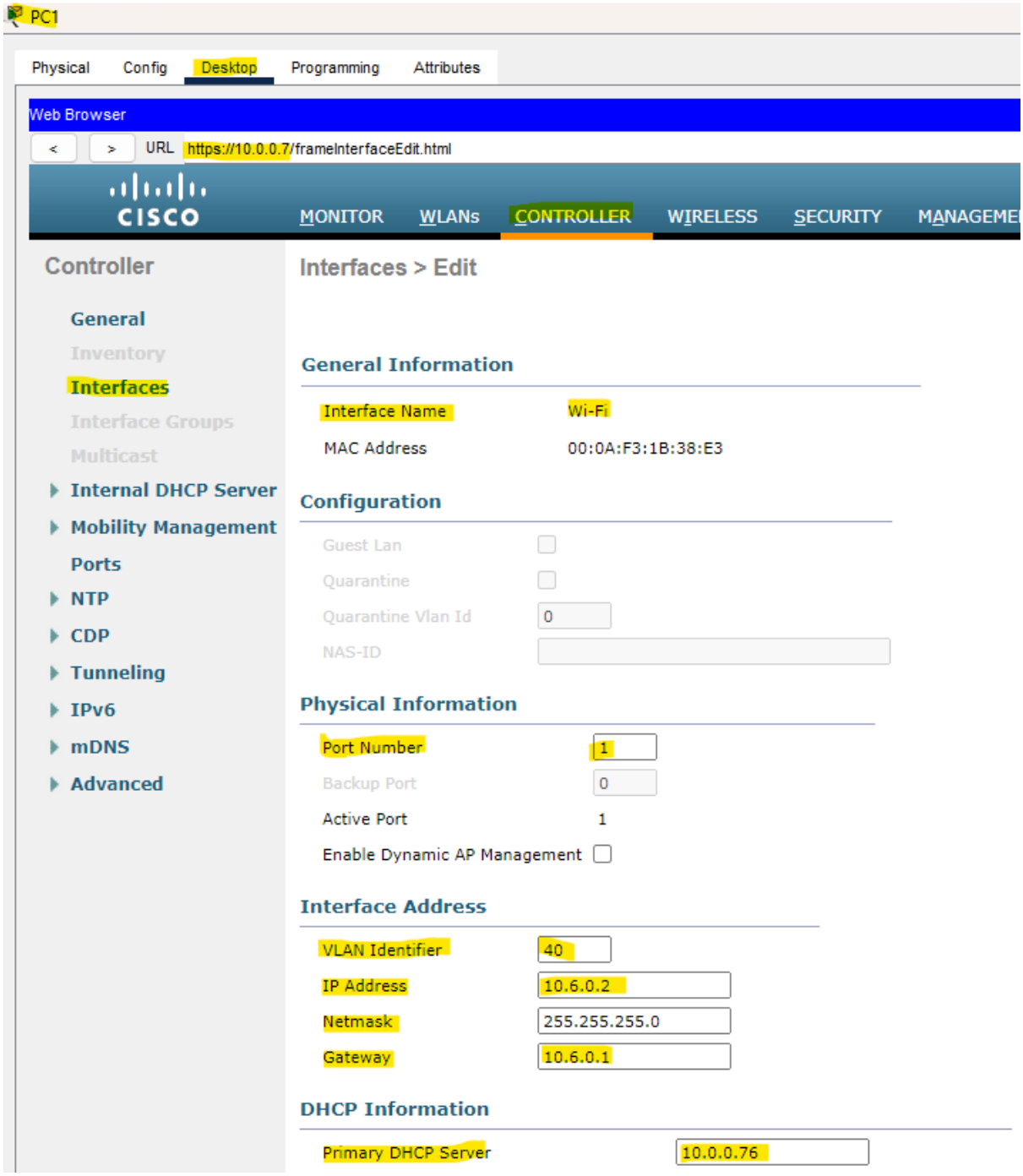

Figure 53. WLC1 GUI access and dynamic interface configuration for the Wi-Fi VLAN

WLAN Configuration

A WLAN profile named Wi-Fi was created on WLC1, with the SSID also set to Wi-Fi, as shown in Figure 54. Under the security settings, WPA2 with AES encryption was enabled, and PSK authentication was configured, as shown in Figure 55. The pre-shared key used for the WLAN was cisco123.

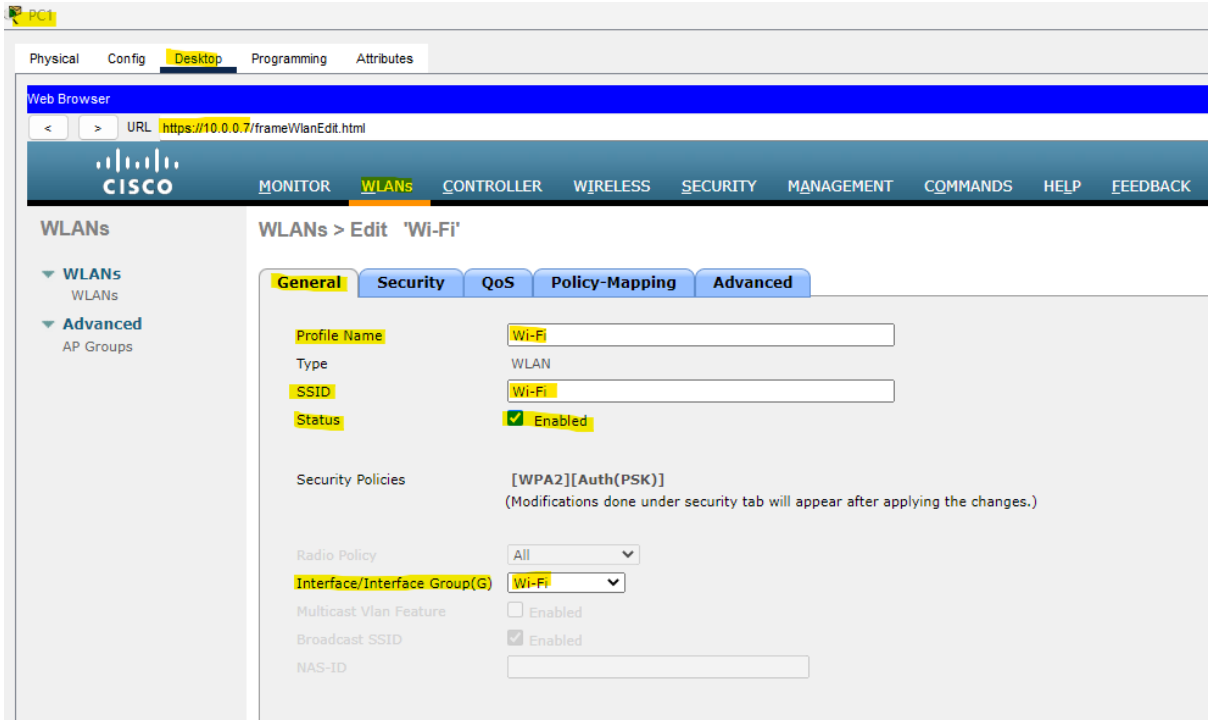

Figure 54. WLAN profile and SSID configuration for Wi-Fi

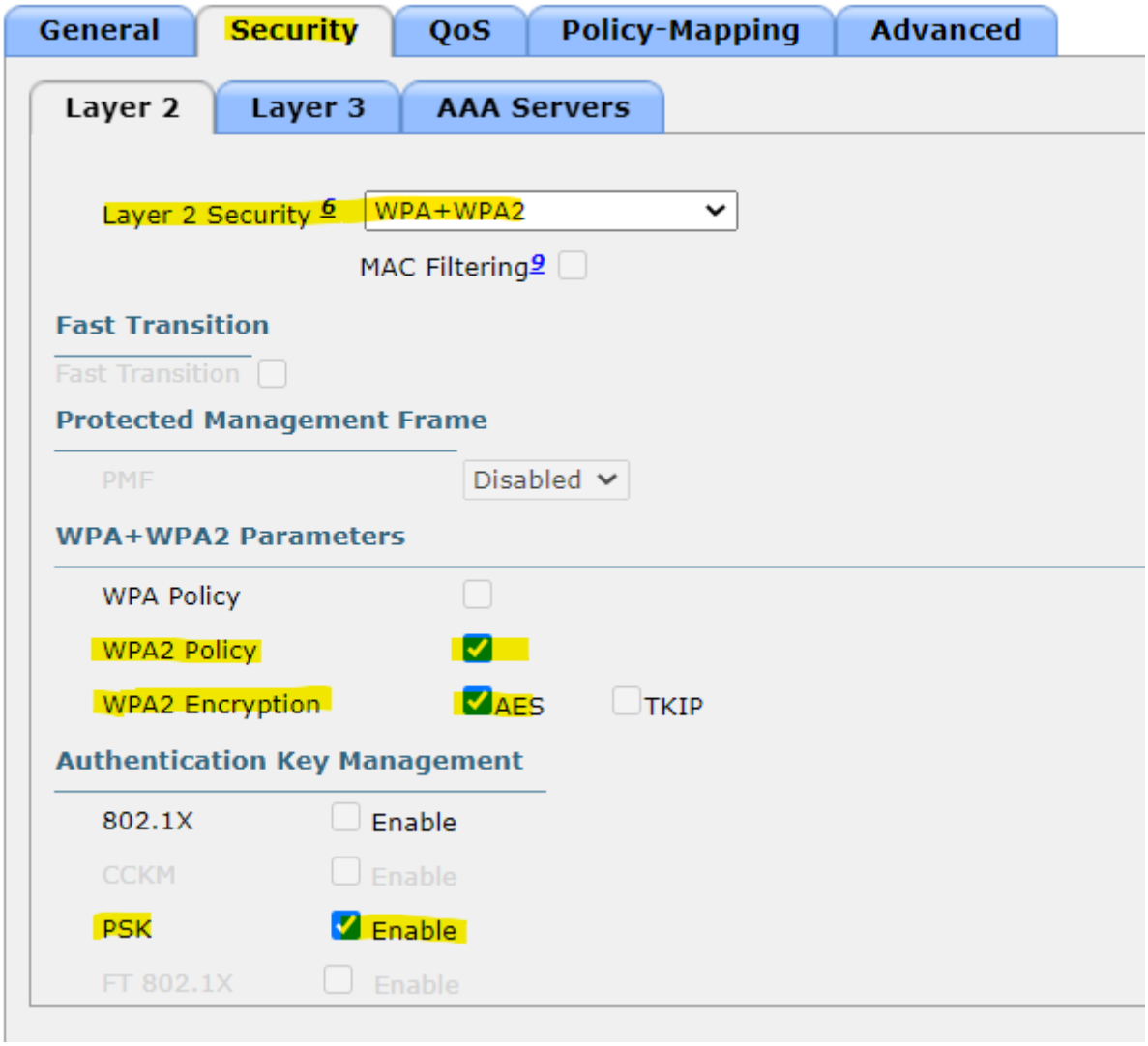

Figure 55. WLAN security configuration using WPA2, AES, and PSK

Wireless Client and LWAP Verification

The wireless client was configured to connect to the Wi-Fi SSID using WPA2-PSK with AES encryption and the password cisco123, as shown in Figure 56. LWAP association was then verified on the WLC, where both LWAP1 and LWAP2 appeared in the access point list, as shown in Figure 57.

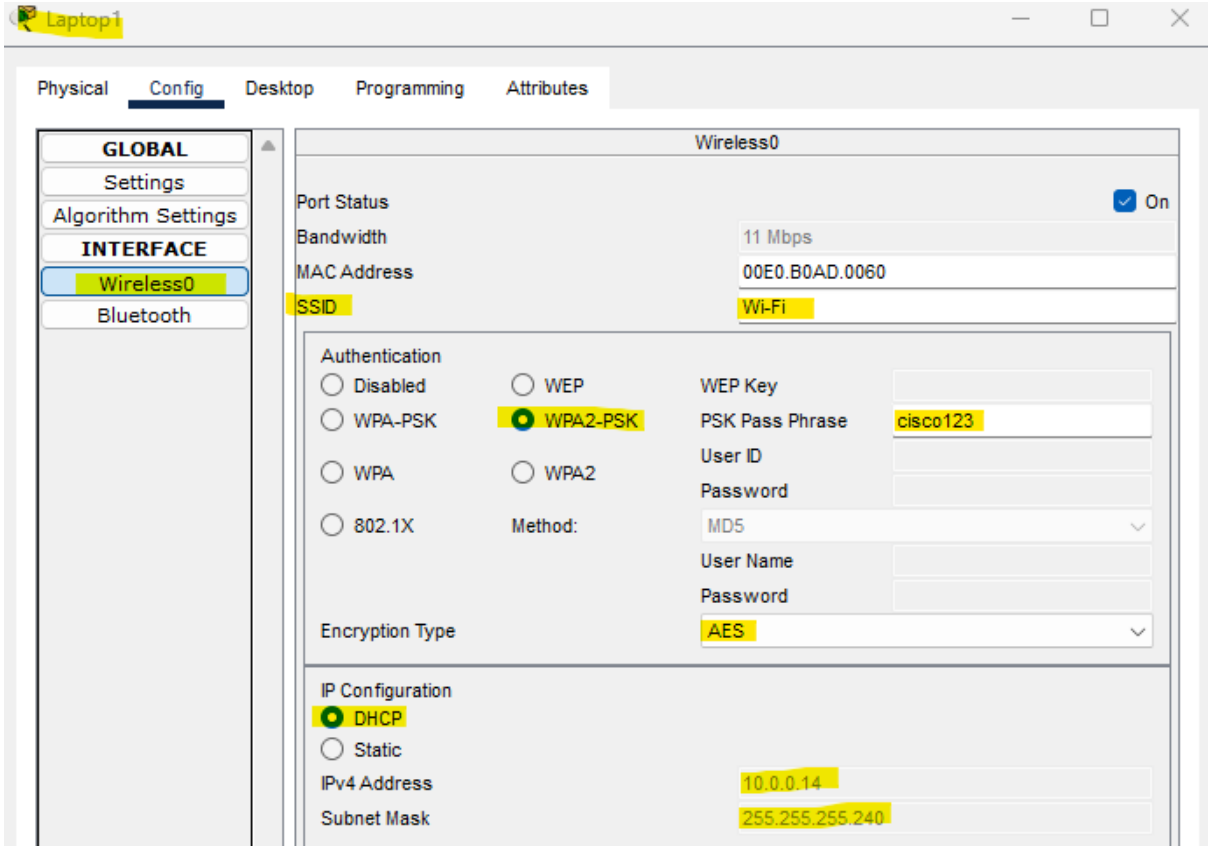

Figure 56. Wireless client configuration for the Wi-Fi SSID

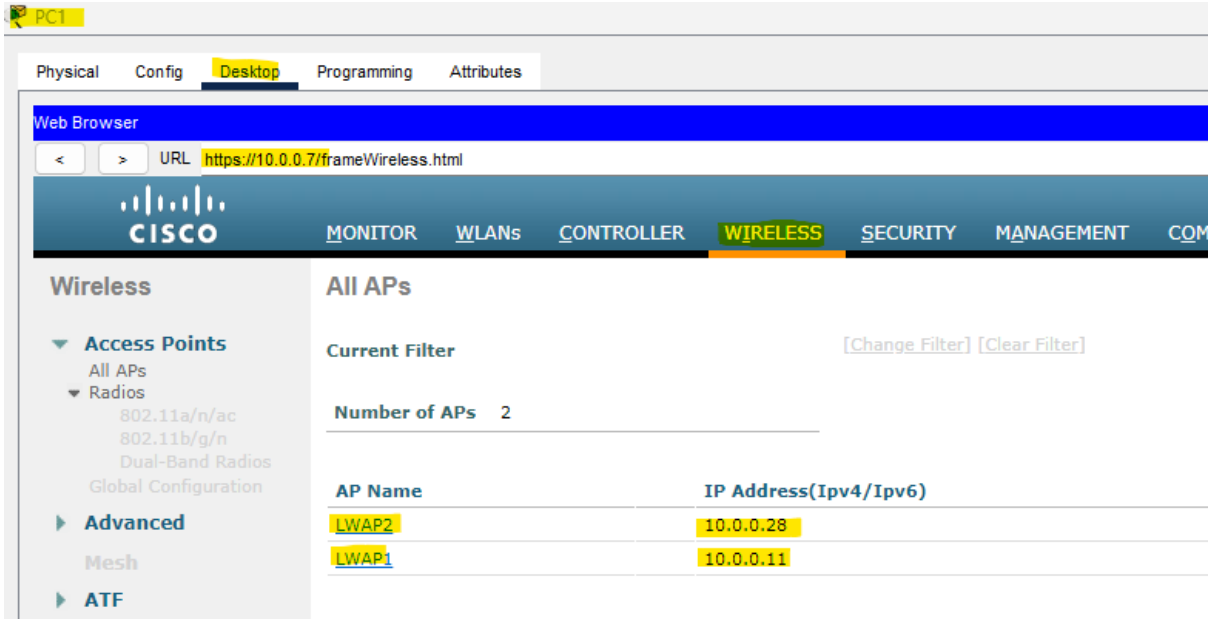

Figure 57. Verification of LWAP association on WLC1